BETÜL TOY · COMPUTER ENGINEER

I turn engineering data into worlds investors can *walk through*.

For solar power plants and residential projects, I build digital twins, photorealistic renders and in-browser 3D experiences from drawings and site data. A project can be seen, explored and understood before it is built.

PROJECTS DELIVERED	PANELS MODELLED	DIGITAL SITE	INTERFACE LANGUAGES
2	38,246	22.20 ha	TR · EN

PROFILE

3D visualization through a computer engineer's eyes

I am a computer engineer, and I approach visualization like a software project: the data is read correctly, the model is built to scale, the experience is tested, and the delivery is documented. That is how a 38,246-panel plant and a two-storey villa both run smoothly in a browser with the same rigour.

Betül TOY

Computer Engineer · 3D Visualization and Digital Twins

01

Data

DWG drawings, layout plans, equipment datasheets and location data are gathered into one source of truth.

DWG · PDF · JSON

02

Model

A to-scale scene is built in Blender; thousands of repeated elements are instanced, and materials and light are tuned to reality.

Blender · Cycles · FBX

03

Experience

Guided tours, datasheets and day/night scenarios are coded in Unity and packaged as WebGL that embeds in any website.

Unity 6 · C# · HLSL · WebGL

04

Delivery

The client receives a complete package: test report, user guide, embed code and editable source projects.

iframe · Next.js · SHA-256

P-01

Serta Solar Power Plant

A digital twin of a solar plant on forested, sloping land that visitors can walk through in the browser.

WHR4+CR · GÖKÇEIÇI / BAYRAMIÇ / ÇANAKKALE

PARTIES	TYPE	DELIVERED	ENGINE	IMAGERY
HTS Solar × CMB Solar	Ground-mounted PV	28 Oct 2026 · Web V2	Unity 6000.4.9f1 · URP · WebGL 2	Blender · 1920×1080



Site overview · 846 panel tables, 10 transformer kiosks

BRIEF

The plant sits on fragmented, sloping land surrounded by forest in Bayramiç, Çanakkale. Investors and partners needed to inspect it as if they were on site, without travelling there.

APPROACH

The site was built as one 3D model tied to the layout plan: panel tables, inverters, transformer kiosks, fencing, gates, walkways and forest. Every repeated element is instanced from a shared model, which lets 38,246 panels and 20,843 trees run smoothly in a browser.

MY ROLE

Site and equipment modelling, the render set, Unity interaction design and code, the WebGL template, the website embed guide and the client delivery package.

OUTCOME

Delivered as a ready-to-host web app, an editable Unity project, a test report and a SHA-256 file manifest. Browser tests logged zero JavaScript or Unity errors.

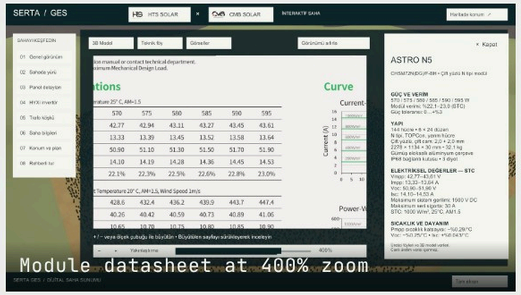
38,246	846	63	10	22.20 ha	2.53 km	20,843
PANELS	PANEL TABLES	INVERTERS	TRANSFORMER KIOSKS	MODEL AREA	WALKING ROUTE	TREE INSTANCES

The interactive app

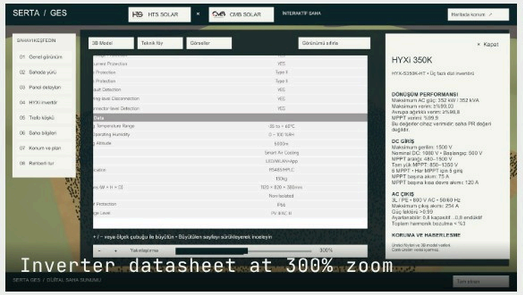
Real screenshots from the app. Visitors walk the route, follow a guided tour at 1×–2× speed, click any equipment and zoom its datasheet up to 500%.



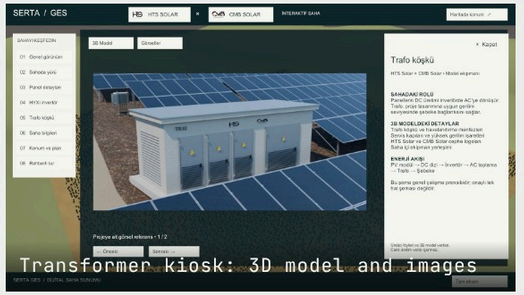
Start screen: route plan and site summary



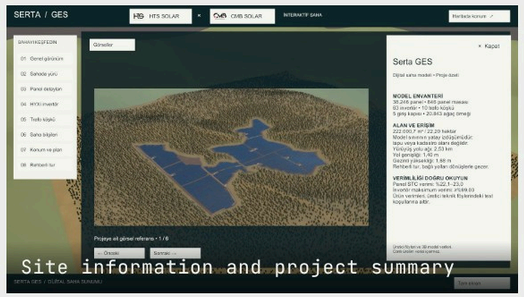
Module datasheet at 400% zoom



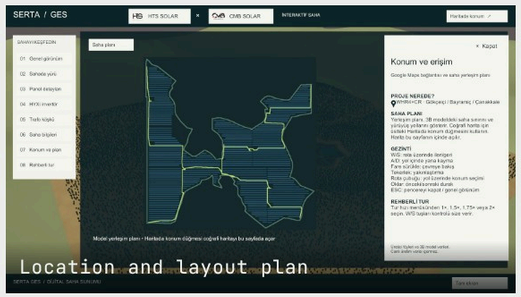
Inverter datasheet at 300% zoom



Transformer kiosk: 3D model and images



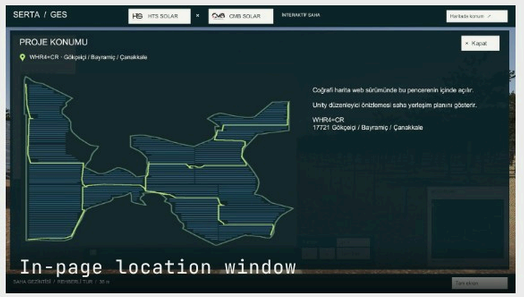
Site information and project summary



Location and layout plan



Guided tour at 2× speed



In-page location window

Module · ASTRO N5

POWER	570–595 W
EFFICIENCY	22.1–23.0% (STC)
CELLS	144 cells, N-type TOPCon, dual glass

Inverter · HYXi 350K

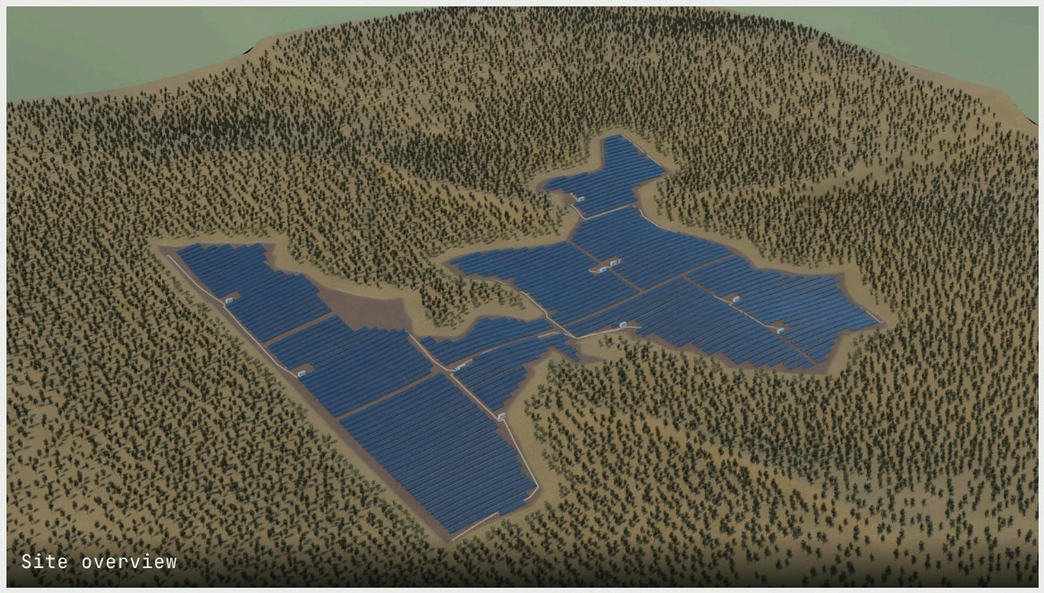
AC POWER	352 kW / 352 kVA
EFFICIENCY	max ≥99.03%
MPPT	6 MPPT, 850–1350 V full load

Energy path

CHAIN	PV module → DC string → Inverter → AC collection → Transformer → Grid
ROUTE	1.40 m wide, 1.68 m eye height

Site and equipment renders

16 images at 1920×1080 from the Blender working model. The last frame delivers the tree asset on a transparent background.


Site overview


Panel rows and transformer kiosk


HYXi string inverter

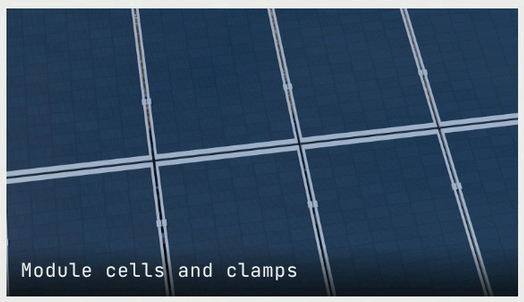


TR10 transformer kiosk with HTS and CMB branding

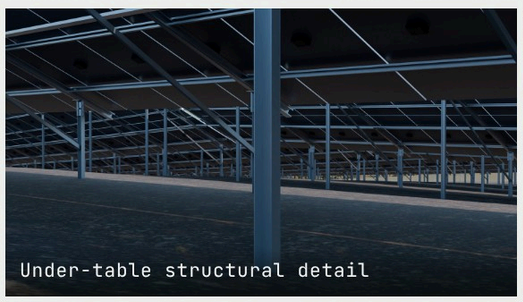

View into the site from the forest edge


Entrance gate and service road


Natural ground and walkway

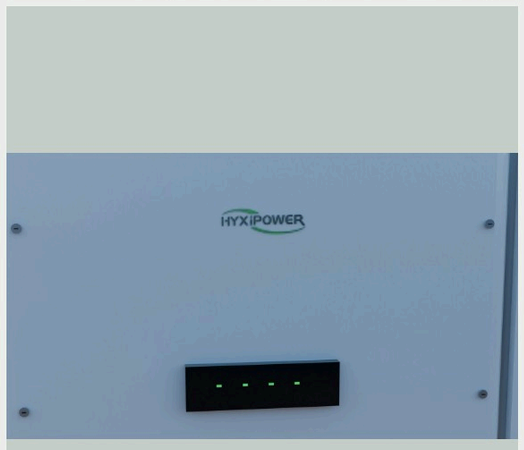

Module cells and clamps


Panel table and mounting structure

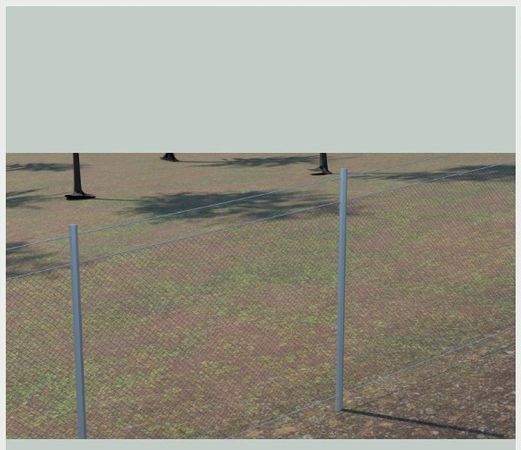

Under-table structural detail



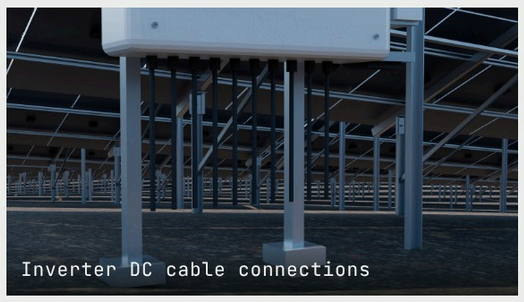
HTS and CMB logos



HYXi logo and status display


Perimeter fence and chain-link mesh


Site plan view from above


Inverter DC cable connections


Tree asset · transparent background

P-02

Twin Villa

Two mirrored, independent 3+1 villas, each with its own pool, garden, upper terrace and parking.

DELIVERED 01.11.2026

PREPARED BY	NET INTERIOR	FACADE WIDTH	RENDER	VIRTUAL TOUR
HTS Solar Enerji A.Ş. by Betül TOY	**101.47 m² / villa**	**15.6 m (both villas)**	**Cycles GPU · 192 spp · AgX**	**Unity 6 · URP · WebGL**



Pool and garden facade

BRIEF

Imagery for sales and presentations that a buyer can understand without reading drawings, plus a virtual tour they can explore on their own.

APPROACH

The architectural DWG became a full Blender 5.2 scene: anthracite brick and mineral render facade, walnut louvres, mosaic pools, travertine paths. The same model moved into Unity 6 as a web tour with day, sunset and night lighting, floor cut-aways and room hotspots.

MY ROLE

Modelling, materials and lighting, an 8-camera render set, landscaping, Unity tour development, the TR/EN interface, FBX export and the delivery document.

OUTCOME

A six-part package: renders, 3D floor plans, the web tour, an FBX model, the Blender source and the Unity project. The tour can be embedded in any website with a copy-paste snippet.

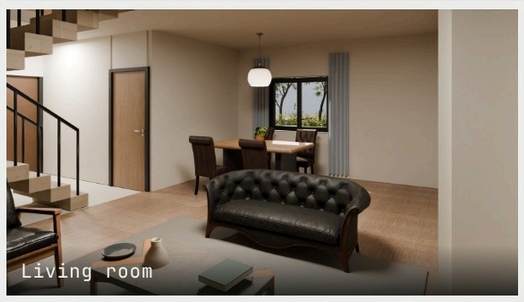
2	101.47 m²	8	~1,600	59,319	3
MIRRORED VILLAS	NET AREA / VILLA	RENDER CAMERAS	ARCHITECTURAL OBJECTS	PLANT INSTANCES	LIGHTING MODES

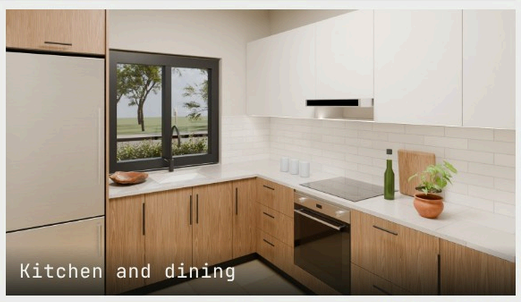
Photorealistic renders

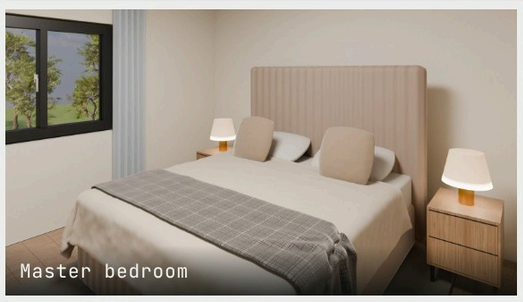
Eight cameras rendered in Blender Cycles at 1920×1080. Click any image to view it full screen.

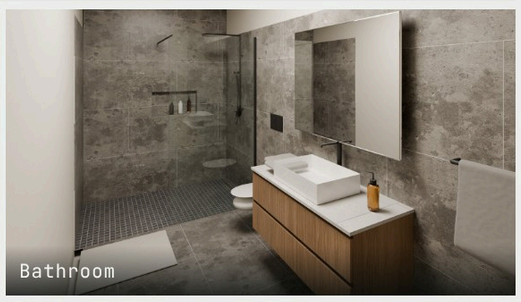

Street facade

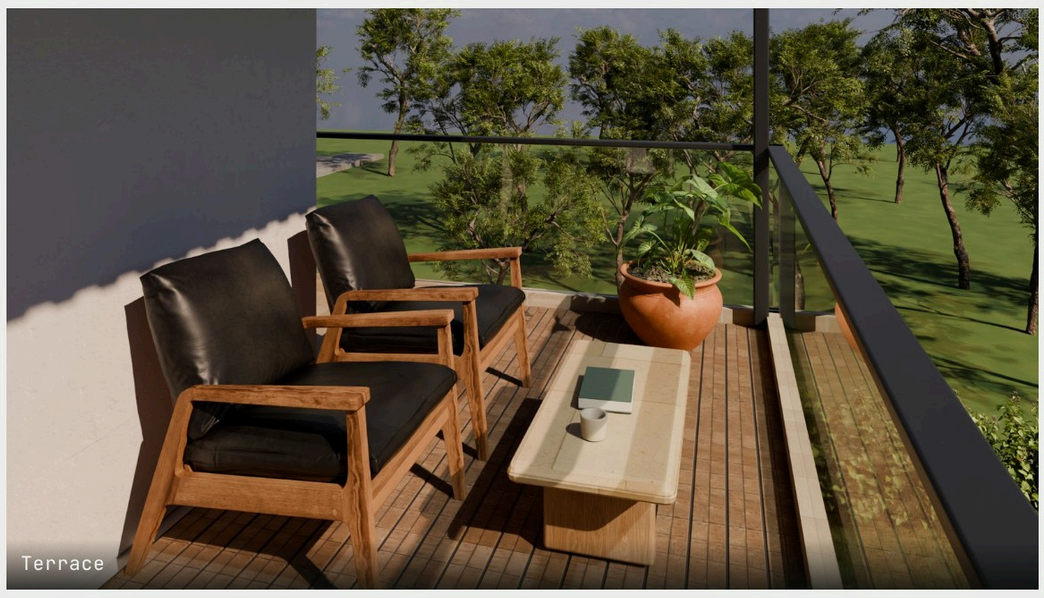

Aerial overview


Living room


Kitchen and dining


Master bedroom


Bathroom


Terrace


Pool and garden

Interactive floor plan

Villa A. Tap a space on the plan or in the list. Areas are net room areas taken from the architectural DWG.



Living & dining	28.97 m²
Kitchen	9.88 m²
Entrance hall	7.31 m²
Guest bathroom	4.76 m²
Garden terrace	13.88 m²
Ground floor total (net)	50.92 m²

Living & dining · 28.97 m²

Oak parquet with dining and seating areas; opens to the garden terrace.

Villa B mirrors Villa A. Villa total (net interior): 101.47 m²; the 13.88 m² garden terrace is not included.

Interactive floor plan

Villa A. Tap a space on the plan or in the list. Areas are net room areas taken from the architectural DWG.



Master bedroom	16.90 m²
Bedroom 2	10.53 m²
Bedroom 3	11.55 m²
Bathroom	8.00 m²
Upper hall	3.57 m²
Upper terrace	—
Upper floor total (net)	50.55 m²

Master bedroom · 16.90 m²

On the garden and pool side, oak parquet.

Villa B mirrors Villa A. Villa total (net interior): 101.47 m²; the 13.88 m² garden terrace is not included.

Virtual tour

Runs in the browser with no install. It loads only when the visitor presses “Start tour”, so the host page stays fast.

SECTIONS	Overview, facade, garden, terrace, ground floor, upper floor, rooms
CUT-AWAYS	Walls are cut so rooms are seen from above
LIGHTING	Day, sunset, night; interior and garden lights respond
ALSO	TR | EN language, ambient sound, screenshot, full screen
WALK-THROUGH	Free walk from the Villa A entrance, stairs included
HOTSPOTS	Room name, area and description
CONTROLS	Desktop WASD + mouse; mobile joystick and pinch zoom
INTEGRATION	iframe or Next.js component; no extra server setup

WHAT I DO

One line from drawing to delivery

Photorealistic visualization

Blender scenes built from DWG and layout plans, with materials, lighting and landscaping, ready for sales and presentations.

Digital twins and site models

3D models that keep large sites light and accurate by instancing tens of thousands of panels and trees.

Interactive web experiences

Guided tours, equipment datasheets, floor cut-aways, day/night lighting and bilingual interfaces in Unity 6 and WebGL.

Delivery engineering

Embed snippets, user guides, test reports, file-integrity manifests and editable source projects.

`Blender 5.2 · Cycles` `Unity 6 · URP` `WebGL 2` `C# · HLSL` `FBX · JSON` `DWG` `iframe · Next.js embed`

CONTACT

Let's walk through your project.

Live 3D tours are shared on request. Write to me about your next plant, residence or presentation.

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